

Shallow water analysis

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1 Shallow water equation

We write the shallow water equation

$$\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} = -g \nabla \eta - \frac{C_f |\mathbf{u}| \mathbf{u}}{h}, \quad (1)$$

$$\partial_t h + \nabla \cdot (\mathbf{u} h) = 0. \quad (2)$$

Neglecting the bottom topography for now, so that $\nabla \eta = \nabla h$. We write it in channel-following coordinate:

$$\partial_t u + u \partial_s u + v \partial_n u + C u v = -g \partial_s h - \frac{C_f |\mathbf{u}|}{h} u, \quad (3)$$

$$\partial_t v + u \partial_s v + v \partial_n v - C u^2 = -g \partial_n h - \frac{C_f |\mathbf{u}|}{h} v, \quad (4)$$

$$\partial_t h + \partial_s (u h) + \partial_n (v h) + C v h = 0. \quad (5)$$

Assume $u = U_0 + u'$, $v = v'$, $C = C'$, $h = H_0 + h'$, and assume that $u'/U_0 \sim h'/H_0 \sim \mathcal{O}(\varepsilon)$. If we neglect all $\mathcal{O}(\varepsilon^2)$ terms, we have

$$\partial_t u' + U_0 \partial_s U_0 + U_0 \partial_s u' + u' \partial_s U_0 + v' \partial_n U_0 = -g \partial_s H_0 - g \partial_s h' - \frac{C_f U_0^2}{H} - \frac{2C_f U_0 u'}{H} + \frac{C_f U_0^2}{H_0^2} h' \quad (6)$$

$$\partial_t v' + U_0 \partial_s v' - C' U_0^2 = -g \partial_n H_0 - g \partial_n h' - \frac{C_f U_0 v'}{H_0} \quad (7)$$

$$\partial_t h' + \partial_s(U_0 H_0) + \partial_s(U_0 h') + \partial_s(u' H_0) + \partial_n(v' H_0) = 0 \quad (8)$$

Leading order balance:

$$U_0 \partial_s U_0 = -g \partial_s H_0 - \frac{C_f U_0^2}{H_0}, \quad (9)$$

$$-g \partial_n H_0 = 0, \quad (10)$$

$$\partial_s(U_0 H_0) = 0. \quad (11)$$

The background state is a bit weird, but let's for now assume we can have a state where $U_0 = U_0 \tilde{\xi}(n)$ and $H_0 = \text{const}$ that can satisfy... We assume a steady state, and non-dimensionalize the $\mathcal{O}(\varepsilon)$ equation with

$$\tilde{u} = u'/U, \tilde{v} = v'/V, \tilde{s} = s/\Lambda, \tilde{n} = n/b, C' = \mathcal{C}\tilde{C}(s), \tilde{h} = h'/\mathcal{H}, \quad (12)$$

we have

$$\frac{UU_0}{\Lambda} \tilde{\xi} \tilde{\partial}_s \tilde{u} + \frac{VU_0}{b} \tilde{v} \tilde{\partial}_n \tilde{\xi} = -\frac{g\mathcal{H}}{\Lambda} \tilde{\partial}_s \tilde{h} - \frac{2C_f U_0 U}{H_0} \tilde{\xi} \tilde{u} + \frac{C_f U_0^2 \mathcal{H}}{H_0^2} \tilde{\xi}^2 \tilde{h}, \quad (13)$$

$$\frac{VU_0}{\Lambda} \tilde{\xi} \tilde{\partial}_s \tilde{v} - \mathcal{C} U_0^2 \tilde{C} \tilde{\xi}^2 = -\frac{g\mathcal{H}}{b} \tilde{\partial}_n \tilde{h} - \frac{C_f U_0 V}{H_0} \tilde{\xi} \tilde{v}, \quad (14)$$

$$\frac{\mathcal{H}U_0}{\Lambda} \tilde{\xi} \tilde{\partial}_s \tilde{h} + \frac{H_0 U}{\Lambda} \tilde{\partial}_s \tilde{u} + \frac{H_0 V}{b} \tilde{\partial}_n \tilde{v} = 0. \quad (15)$$

Define

$$\alpha = b/\Lambda, \gamma = V/U, \varepsilon_1 = U/U_0, \varepsilon_2 = \mathcal{H}/H_0, Fr = U_0/\sqrt{gH_0}, F_c = C_f \Lambda/H_0, Ci = \mathcal{C}b \quad (16)$$

we have

$$\tilde{\xi} \tilde{\partial}_s \tilde{u} + \frac{\gamma}{\alpha} \tilde{v} \tilde{\partial}_n \tilde{\xi} = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_s \tilde{h} - 2F_c \tilde{\xi} \tilde{u} + F_c \frac{\varepsilon_2}{\varepsilon_1} \tilde{\xi}^2 \tilde{h}, \quad (17)$$

$$\tilde{\xi} \tilde{\partial}_s \tilde{v} - \frac{Ci}{\varepsilon_1 \alpha \gamma} \tilde{C} \tilde{\xi}^2 = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2 \alpha \gamma} \tilde{\partial}_n \tilde{h} - F_c \tilde{\xi} \tilde{v}, \quad (18)$$

$$\tilde{\partial}_s \tilde{u} + \frac{\gamma}{\alpha} \tilde{\partial}_n \tilde{v} + \frac{\varepsilon_2}{\varepsilon_1} \tilde{\xi} \tilde{\partial}_s \tilde{h} = 0. \quad (19)$$

We typically assume $\alpha = \gamma$. **If we assume** $Ci \sim \varepsilon_1 \sim \varepsilon_2 \sim \mathcal{O}(\varepsilon)$, $\alpha \sim \mathcal{O}(\varepsilon^{1/2})$, $Fr \sim F_c \sim 1$, the leading order equation becomes

$$\tilde{\xi} \tilde{\partial}_s \tilde{u} + \tilde{v} \tilde{\partial}_n \tilde{\xi} = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_s \tilde{h} - 2F_c \tilde{\xi} \tilde{u} + F_c \frac{\varepsilon_2}{\varepsilon_1} \tilde{\xi}^2 \tilde{h}, \quad (20)$$

$$\frac{1}{\varepsilon_1 \alpha^2} Ci \tilde{C} \tilde{\xi}^2 = \frac{1}{\varepsilon_1 \alpha^2} \frac{\varepsilon_2}{Fr^2} \tilde{\partial}_n \tilde{h}, \quad (21)$$

$$\tilde{\partial}_s \tilde{u} + \tilde{\partial}_n \tilde{v} + \frac{\varepsilon_2}{\varepsilon_1} \tilde{\xi} \tilde{\partial}_s \tilde{h} = 0. \quad (22)$$

Equations 20 and Equation 21 lead to the solution in Ikeda et al.

If we assume that $\varepsilon_2 \sim \mathcal{O}(\varepsilon^2)$, $Ci \sim \varepsilon_1 \sim Fr^2 \sim \mathcal{O}(\varepsilon)$, **and set** $\alpha = 1$, the leading order equation becomes

$$\tilde{\xi} \tilde{\partial}_s \tilde{u} + \tilde{v} \tilde{\partial}_n \tilde{\xi} = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_s \tilde{h} - 2F_c \tilde{\xi} \tilde{u}, \quad (23)$$

$$\tilde{\xi} \tilde{\partial}_s \tilde{v} - \frac{Ci}{\varepsilon_1 \alpha^2} \tilde{C} \tilde{\xi}^2 = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_n \tilde{h} - F_c \tilde{\xi} \tilde{v}, \quad (24)$$

$$\tilde{\partial}_s \tilde{u} + \tilde{\partial}_n \tilde{v} = 0. \quad (25)$$

Equations 23-25 leads to the QGPV equation

$$\tilde{\xi} \tilde{\partial}_s \left(\tilde{\nabla}^2 \tilde{\psi} - \frac{Ci}{\varepsilon_1 \alpha^2} \tilde{C} \tilde{\xi} \right) + \tilde{\beta} \tilde{v} = -F_c \tilde{\xi} \tilde{\nabla}^2 \tilde{\psi} + 2F_c \tilde{u} \tilde{\partial}_n \tilde{\xi} + F_c \tilde{\xi} \tilde{\partial}_n \tilde{u}, \quad (26)$$

where

$$\tilde{\beta} = -\tilde{\partial}_{nn} \tilde{\xi}, \quad \tilde{\nabla}^2 \tilde{\psi} = \tilde{\partial}_s \tilde{v} - \tilde{\partial}_n \tilde{u}. \quad (27)$$

We can put:

$$\tilde{\xi}(\tilde{n}) \tilde{\partial}_s \tilde{u} + \tilde{v} \tilde{\partial}_n \tilde{\xi}(\tilde{n}) + \frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_s \tilde{h} + 2F_c \tilde{\xi}(\tilde{n}) \tilde{u} - F_c \frac{\varepsilon_2}{\varepsilon_1} \tilde{\xi}(\tilde{n})^2 \tilde{h} = 0, \quad (28)$$

$$\tilde{\xi}(\tilde{n}) \tilde{\partial}_s \tilde{v} - \frac{Ci}{\varepsilon_1 \alpha^2} \tilde{C}(\tilde{s}) \tilde{\xi}(\tilde{n})^2 + \frac{\varepsilon_2}{\varepsilon_1 Fr^2 \alpha^2} \tilde{\partial}_n \tilde{h} + F_c \tilde{\xi}(\tilde{n}) \tilde{v} = 0, \quad (29)$$

$$\tilde{\partial}_s \tilde{u} + \tilde{\partial}_n \tilde{v} + \frac{\varepsilon_2}{\varepsilon_1} \tilde{\xi}(\tilde{n}) \tilde{\partial}_s \tilde{h} = 0. \quad (30)$$